PH101T: Quantum Physics and Applications: Problems

Bring your papers
Simple calculator allowed.
No handwritten papers or electronic gadget - other than to read questions
Invigilators will penalise for discussions during the test
or for refering to answers in mobile

Maximum marks:20. Weightage will be decided later.
Show the expressions and numbers asked or graph separately
(for ease of evaluation)

1. One Dimensional Coulomb Potential: Schrodinger Wave function:
$\Psi(x) = 2\, b^{\frac{3}{2}}\, x e^{-bx}$. $x > 0$.

a. Find the expectation value for $< \frac{1}{x^2} >$

b. Find the Expectation value for Kinetic Energy (mass=m) . (3 mark)

2. The wavefunction for a particle of mass m (nonrelativistic Schrodinger equation) is given by
$\Psi(x) = \sqrt{b}e^{-bx}$ for $x > 0$; $\Psi(x) = \sqrt{b}e^{+bx}$ for $x < 0$.   b is constant.

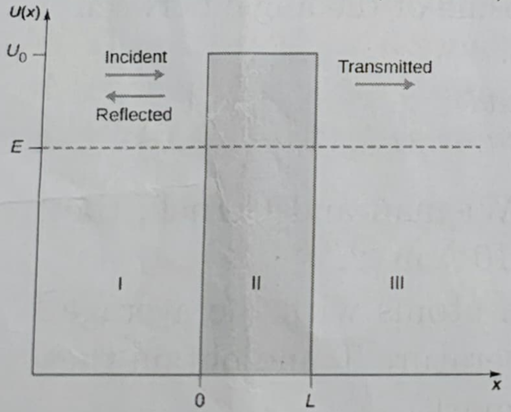
a. Find the expectation value of the Kinetic Energy of the particle.

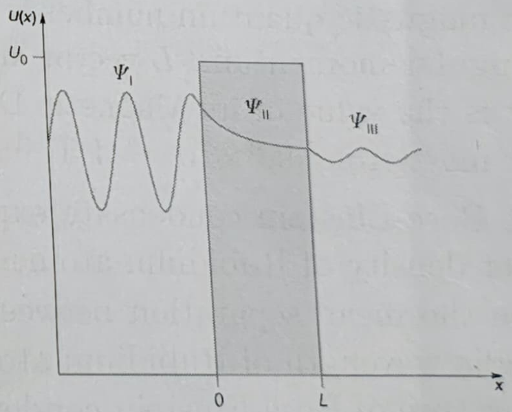
b. Plot a graph of x versus Potential. (3 mark)
[Think of boundary conditions at x=0. Think of the constant occuring in the potential.]

3. A neutron of mc^2= 940 MeV is confined to the nuclear size of 1.2 fm.
hc = 1240 MeV fm (fm=femtometre = 10^{-15} m.)
Assume non-relativistic Schrodinger equation for a free particle with ∞ barrier confining it.

- Find the Ground State Energy. (3 mark)

4. Consider the Potential and the Wavefunction for a beam of particles incident from the left (-x side) as shown below.





a. Write the wavefunction in the three regions for the sketch.
Choose your favourite value for wavenumber.

b. There are 6 undetermined constants. Which of them is/are equated to zero? (3 mark)

5. Consider the Fermi-Dirac distribution of electron gas in a metal as function of energy.
Fermi-Dirac Distribution at Temperature T: $p(E) = \frac{1}{e^{\frac{E-E_F}{k_B T}}+1}$

Density of States (per Unit Volume) $8\pi\sqrt{2}\left(\frac{mc^2}{h^2}\right)^{\frac{3}{2}}\sqrt{E}\,dE$
Hence,
Number of electrons per unit energy n(E) in a volume V:

$$n(E)dE = 8\pi V\sqrt{2}\left(\frac{m}{h^2}\right)^{\frac{3}{2}} \frac{\sqrt{E}dE}{e^{\frac{E-E_F}{k_B T}}+1}$$

where m=mass of electron, T=Temperature, k_B:Boltzmann constant
c=speed of light in vacuum, h=Planck constant
Fermi Energy E_F depends only on the Number Density of Electrons and not on Temperature.

- Write the **Relation between Electron number density and Fermi Energy**. (3 mark)
[Do not try to directly integrate the above expression for n(E) as a function of Temperature].

6. Consider non-Relativistic Schrodinger Equation for a particle of mass m in a Potential given by
$V = \infty$ for $x < 0$; $V = -V_o$ for $0 < x < L$. V=0 for $x > L$.
Find the minimum V_o for which there exists at least one Bound State
[for which Energy Eigenvalue < 0; wavefunction vanishes at ∞]
You can try to solve exactly and check the minimum wavenumber k for which tan(kL) is negative. OR
Assume that Energy is very close to but slightly less than 0. (3 mark)

7. An electron is in an angular momentum state ℓ=3.
a. What is the value of its angular momentum?
b. If the magnetic quantum number is 3, what is cosine of the angle between the angular momentum, L vector and the z-axis?
c. What is the value of its Magnetic Dipole Moment?
(Bohr magneton=9.3×10^{-24} J T^{-1} (2 mark)

8. In the Bose-Einstein condensate experiment of Wieman and Cornell, the number density of Rubidium atoms was n = 5×10^{18} m^{-3}.
Equate the mean separation between Rubedium atoms with the average deBroglie wavength of Rubidium atoms at Temperature T and obtain the Temperature of Bose-Einstein condensation. (3 mark)
Mass of Rubidium atom $\sim 1.44\times10^{-25}$ kg.
$k_B \sim 1.38\times10^{-23}$ m^2 kg s^{-2} K^{-1}.

Kaustav

PH401T Quantum Mechanics: Final Exam 2025–26: Batch S2

This is only for Batch S2 of PH101T (2025)

Write your Name and Roll Number.

Your maximum mark for this exam will be 45.

You are allowed to bring your Notes in paper. But No Electronic Gadgets, No Internet, No Discussion.
Simple Calculator is allowed.

Present the **essential Expression, numbers with units Seperately** , *in a way they are easily noticeable.*
Draw **Graphs** with **proper X-** and **Y-axes, suitable Caption** and name the curves properly (with arrows, if relevant).
Generally it is a good idea to define the variables used in the expressions.

I do not expect exact answer to any question involving numbers. Sufficient if you get the number with 10% error

There are questions on **Both Sides** *of this question paper.*

Value of some constants you might need:
Planck's constant $h \sim 6.626\times10^{-34}$ Joule sec.
$\hbar$ c = 197 MeV fm (fm=femtometre = 10^{-15} m.)
Speed of light in vacuum $c \sim 3\times10^{8}$ metre/sec
Boltzmann constant $k_B \sim 1.38\times10^{-23}$ Joule/Kelvin

1. For a quantum mechanical particle trapped between x=0 and x=L by an ∞ Barrier Potential, **show** that the quantum mechanical wavefunctions for various energy levels, $\Psi_n(x)$, n=1,2,... **form an orthonormal basis.** "*Orthonormal*" requires two conditions. Show the Expression for each condition and the final answer. (4 mark)

2. A one dimensional harmonic oscillator Potential energy function is $U(x) = \frac{1}{2}m\omega^2x^2$, where m is the mass and ω is a constant.
a. Write the Time-Independent Schrodinger equation for this system. The Ground State has Wavefunction $\Psi(x) = Ae^{-\alpha x^2}$.
b. Determine α by comparing the terms with x^2 (or any method).
c. Determine A by normalising the wavefunction.
d. Plot x versus the Potential for $m\omega^2$=1, between x=±1.
e. Plot the Ground state wavefunction in the same graph. (6 mark)

3. A particle of mass m satisfying non-relativistic Schrodinger equation is trapped by Infinity barrier at $x = \pm L$
a. Write the **normalised ground state** wavefunction.
b. Find the uncertainty in its position Δx.

For (b), I shall check (i) expression used to derive the Uncertainty. (ii) value of Δx. Write them clearly, separately. (3 mark)

4. Consider the one dimensional Schrodinger equation for a particle of mass m for which the solution is given by $\Psi(x) = 2\, b^{\frac{3}{2}}\, x e^{-bx}$ for $x > 0$
$\Psi(x) = 0$ for $x < 0$.

4a. Determine the value of **x** for which the Probability Density is maximum.

4b. Find the expectation value for $< \frac{1}{x^2} >$.

An **electron** is in an angular momentum state ℓ=2.

4c. Write the expression for its its angular momentum.

4d. If the magnetic quantum number is 1, what is cosine of the angle between the angular momentum, L vector and the z-axis? (4 mark)

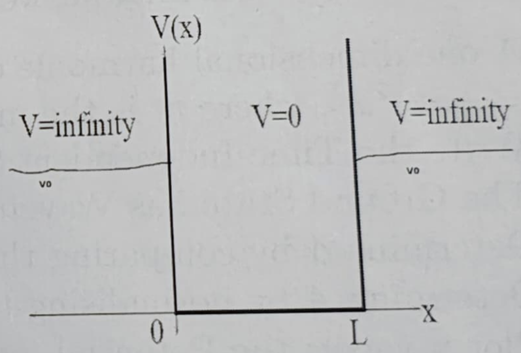
5. If $<p>$ is the expectation value of the Momentum Operator for the Normalised State $\Psi(x)$, determine the expectation value of the momentum operator for the state $\Psi_1(x) = e^{-i\frac{p_0 x}{\hbar}}\Psi(x)$ (2 mark)

6. Consider two orthonormal eigenfunctions of a quantum mechanical system corresponding to two different Energy Eigenvalues E_1 and E_2, given by the stationary state soltuions $|u_1>$ and $|u_2>$.
At time t=0, the system is in a state given by the wavefunction
$|\Psi(0)> = \beta\,(|u_1> - |u_2>)$

6a. Determine the constant β by Normalising $|\Psi(0)>$.
$\Psi(t)$ will evolve with time because of the time dependent part which depends on the Energy Eigenvalue for each term of the expression.

6b. Determine the shortest time t_0 at which $|\Psi(t)>$ will be normal to $|\Psi(0)>$ (5 mark)

7. Consider one-dimensional Schrodinger equation for a particle of mass m in a Potential $V(x)$ shown here.



(i.) V = 0 for $0 < x < L$.
V=∞: $x < 0$ or $x > L$
(Infinity Barrier)

(ii) V=0 for $0 < x < L$;
V = V0 outside (Finite Barrier).
[Thin curve Potential]

a. Write the Normalised Ground State Wavefunction for case(i.).

b. Draw a graph of the above wavefunction (x versus amplitude) in **continuous curve.**

b. Draw a graph of the Ground State wavefunction, **overplotting the above curve with dashed line** for the

Potential of case (ii) - finite barrier.
This graph should be qualitatively correct (but no need to normalise exactly). (4 mark)

8. Assume that neutral π^0 (pion) of mass 135 $\frac{MeV}{c^2}$ [c=speed of light] is confined to the nuclear size of 0.7 fm.
Assume non-relativistic Schrodinger equation for a free particle with ∞ barrier confining it.

- Find the Ground State Energy.
Pion life time=8.5×10^{-17}seconds. $\tau c \sim 25$ nm.
- Use Heisenberg's principle to estimate the Uncertainty in its Energy.
(4 mark)

9. Consider the Fermi-Dirac distribution of electron gas in a metal as function of energy.
Number of electrons per unit energy n(E) in a volume V:

$$n(E)dE = 8\pi V\sqrt{2}\left(\frac{m}{h^2}\right)^{\frac{3}{2}} \frac{\sqrt{E}dE}{e^{\frac{E-E_F}{kT}}+1}$$

where m=mass of electron, T=Temperature, k:Boltzmann constant
c=speed of light in vacuum, h=Planck constant
Fermi Energy E_F depends only on the Number Density of Electrons.

- Find the Fermi Energy (in eV) of Conduction Band Electrons in a metal which has 1.25×10^{29} Electrons per cubic metre. (3 mark)

$\frac{h^2}{2m}\left(\frac{3}{\pi}\right)^{\frac{2}{3}} \sim 0.3\ (\text{nm})^2$ eV

10. Argon Gas at Temperature T=293 K in a container obeys Maxwell-Boltzmann statistics.
Its Density of States is given by $g(E) = \frac{2N}{\sqrt{\pi}(kT)^{\frac{3}{2}}}\sqrt{E}$, where E=energy of the atom, N=Number density of atoms and k=Boltzmann constant.
k T = 0.0252 eV for this gas.
a. Write the Expression for Energy Distribution for this gas.
b. What fraction of Argons in this system are at Energy between 0.025074 eV and 0.025326 eV? (3 mark)

11. In a Bose-Einstein condensate experiment, the number density of Rubidium atoms was n = 8×10^{18} m^{-3}.
a. At a temperature T, write the expression for its mean magnitude of Momentum due to thermal motion.

b. Equate the mean separation between Rubidium atoms with their average deBroglie wavength at Temperature T and obtain the Temperature of Bose-Einstein condensation. (3 mark)
Mass of Rubidium atom $\sim 1.44\times10^{-25}$ kg.
$\frac{h}{\sqrt{3km}} \sim 0.25$ nm $K^{-\frac{1}{2}}$.

12. Consider a one-dimensional crystal with lattice parameter a [mean separation between centres of atoms] and Mass of atom M.
The displacement of the nth atom about its mean, u_n satisfies the equation

$$M\frac{d^2u_n}{dt^2} = C\left(u_{n+1} - 2u_n + u_{n-1}\right)$$

where C is the force constant.
The soluton is $u_n(x,t) = Ae^{-i(kna-\omega t)}$ where ω is the angular frequency of the mode.

a. Write the Dispersion relation [relation between ω and k].
b. What is the Group Velocity of this mode?
c. Plot a graph of ω as a function of k for k in the range of $-\frac{\pi}{a}$ to $+\frac{\pi}{a}$ (6 mark)